

KNOXVILLE, IA, USA

WMO: 725493

Lat: 41.299N

Lon: 93.114W

Elev: 282

StdP: 97.98

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-17.0	-24.9	0.4	-19.1	-22.3	0.5	-15.8	12.6	-4.9	11.6	-3.5	3.9	320	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	33.9	25.2	32.4	24.8	31.1	24.1	27.1	32.0	26.0	30.6	25.1	29.4	4.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.1	30.9	24.8	20.5	29.6	23.4	18.8	28.3	87.9	32.2	82.2	30.6	78.0	29.3	37.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.4	8.4	DB	-24.8	36.0	4.4	3.1	-27.9	38.2	-30.5	40.0	-32.9	41.8	-36.1	44.0	(4)
(5)			WB	-25.0	28.1	4.3	1.2	-28.0	29.0	-30.5	29.8	-32.9	30.5	-36.0	31.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-3.9	-2.3	4.4	11.3	17.3	22.3	24.5	23.3	19.4	12.1	5.5	-1.1	(6)
(7)		DBStd	11.38	6.88	7.19	6.95	5.24	4.62	3.49	3.29	3.11	4.64	5.34	6.62	6.29	(7)
(8)		HDD10.0	1576	432	347	199	45	3	0	0	0	1	39	163	346	(8)
(9)		HDD18.3	3244	690	578	433	216	77	8	1	2	42	204	391	602	(9)
(10)		CDD10.0	1994	0	2	27	85	228	371	451	413	284	104	27	2	(10)
(11)		CDD18.3	621	0	0	1	6	43	128	193	157	76	11	6	0	(11)
(12)		CDH23.3	5709	0	0	10	77	372	1127	1923	1367	648	87	98	0	(12)
(13)	CDH26.7	2077	0	0	1	13	97	395	783	486	216	16	70	0	(13)	
(14)	Wind	WSAvg	3.9	4.4	4.3	4.6	4.7	4.1	3.5	2.9	2.7	3.3	4.0	4.3	4.3	(14)
(15)	Precipitation	PrecAvg	917	26	33	52	98	125	138	109	104	89	61	48	34	(15)
(16)		PrecMax	1344	57	62	137	204	243	329	315	253	184	154	124	120	(16)
(17)		PrecMin	594	14	6	1	40	30	25	26	3	42	8	1	2	(17)
(18)		PrecStd	184	9	15	34	38	52	73	69	67	41	39	30	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	18.6	24.8	28.0	32.3	34.0	37.4	34.9	33.6	28.7	39.2	17.4	(19)
(20)			MCWB	8.5	11.9	16.7	18.7	23.1	25.2	25.8	24.7	24.8	21.3	28.9	13.8	(20)
(21)		2%	DB	10.2	14.8	21.3	25.9	29.0	32.4	34.2	32.7	31.3	26.0	22.0	12.7	(21)
(22)			MCWB	6.1	9.3	14.3	16.8	21.2	24.3	25.9	25.0	22.9	18.4	14.9	9.1	(22)
(23)		5%	DB	7.4	10.9	17.8	22.7	27.3	30.9	32.5	31.2	29.0	22.9	17.4	9.8	(23)
(24)			MCWB	4.3	7.0	12.7	15.2	19.6	23.8	25.5	24.6	22.4	17.0	12.2	7.2	(24)
(25)		10%	DB	4.8	7.5	15.0	20.2	25.1	28.9	31.0	29.1	27.4	21.0	14.0	7.3	(25)
(26)			MCWB	2.3	4.5	10.1	13.6	18.6	22.8	24.7	23.4	21.1	15.8	9.6	4.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	13.9	18.0	20.6	24.7	27.0	29.2	28.1	25.9	22.9	27.3	14.5	(27)
(28)			MCDWB	12.6	17.3	22.8	26.6	29.8	31.9	35.0	33.0	30.4	27.6	37.5	17.1	(28)
(29)		2%	WB	6.4	9.6	15.8	18.2	22.6	25.7	27.7	26.6	24.6	20.4	15.5	10.4	(29)
(30)			MCDWB	9.9	14.1	19.5	23.4	27.8	30.8	32.7	31.1	29.4	24.0	20.1	11.8	(30)
(31)		5%	WB	4.4	7.0	13.6	16.2	21.0	24.6	26.4	25.4	23.3	18.6	12.9	6.8	(31)
(32)			MCDWB	7.5	10.5	17.7	21.5	25.7	29.2	30.9	29.4	27.6	21.9	17.0	9.0	(32)
(33)		10%	WB	2.4	4.6	10.3	14.6	19.6	23.5	25.3	24.2	22.1	16.3	10.2	4.5	(33)
(34)			MCDWB	4.6	7.6	14.5	19.3	24.0	27.9	29.5	28.3	26.2	20.0	13.9	7.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	9.0	10.0	11.2	10.4	9.9	10.0	9.9	11.3	10.5	9.8	8.1	(35)
(36)			MCDBR	12.1	13.3	14.5	15.2	12.5	11.2	11.3	11.3	12.3	12.8	14.4	10.9	(36)
(37)			MCWBR	9.0	9.2	8.9	8.8	6.6	6.0	5.7	6.0	6.1	7.3	9.5	8.3	(37)
(38)		5% WB	MCDBR	11.8	11.8	12.5	13.1	11.0	10.1	10.2	9.7	10.5	10.7	13.4	9.9	(38)
(39)			MCWBR	8.9	8.8	8.4	8.9	6.8	6.4	6.3	6.3	5.8	7.8	9.7	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.314	0.333	0.386	0.409	0.420	0.436	0.419	0.383	0.341	0.300	0.290	(40)	
(41)		taud	2.428	2.353	2.394	2.262	2.266	2.281	2.225	2.292	2.342	2.459	2.504	2.439	(41)	
(42)		Ebn at Noon	872	894	917	884	868	857	838	843	850	854	852	836	(42)	
(43)		Edh at Noon	78	98	106	131	133	132	138	125	111	88	72	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.92	2.91	3.78	4.75	5.59	6.21	6.26	5.44	4.62	3.10	2.11	1.62	(44)	
(45)		RadStd	0.18	0.21	0.34	0.35	0.43	0.61	0.46	0.33	0.34	0.35	0.18	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page